Zacharie Makadai

M.Sc. in Fundamental Sciences
M.Sc. in Applied Mathematics
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PROFILE SUMMARY

A highly motivated mathematics graduate with a strong background in **PDEs, Numerical Methods for PDEs,Sobolev Space,Algorithm and Graph theory, Coding theory,Cryptography, Riemannian Geometry, ...**. Gained extensive experience in Numerical simulation and Statistical tools such as **Python(numpy,networkx,..), R,Sagemath and Scilab** through rigorous coursework at the African Institute for Mathematical Sciences (AIMS). Graduated with First-Class Honors from University of Maroua in Cameroon, where leadership roles, honed my teamwork, organization, and problem-solving skills. Passionate about applying mathematical techniques and seeking a PhD opportunity for Applied mathematics .

EDUCATION

- **Young African Mathematician (YAM) Fellow**, Hausdorff Center for Mathematics, University of Bonn

  Conducted research at the Institute for Numerical Simulation (INS) under the mentorship of Prof. Dr. Joscha Gedicke. Oct 2026-Current

- Msc **in Mathematical Science with distinction (GPA: 3.68/4.0)**
  AIMS-Cameroon, Oct. 2025 – Jun. 2026

- **Msc. in Applied Mathematics with First-Class Honors (GPA: 3.36/4.0).**
  Faculty of Science, University of Maroua, sept . 2023 – sept. 2025
  Specialize at Numerical analysis

- **Bsc. in Mathematics and Computer Science (GPA: 2.87/4.0)**
  Faculty of Science, University of Maroua Sept. 2019 - Jun. 2022

RESEARCH INTERESTS

- Numerical method and simulation apply on fluid dynamics
- Sobolev Space, and the related other Spaces
- Investigate the New numerical methods for solving a PDEs as XFEM (eXtended Finite Element Method),VEM (Virtual Element Method),...
- Applied mathematics in security using Cryptography

RESEARCH EXPERIENCE

Preprint (submitted)
A Non-Adaptive FASTEST Finite Volume Method for a 2D Keller–Segel System
Co-authors: A.C. Kengni Jotsa, J. Tagoudjeu, M. Zagour
Submitted to: Computers and Mathematics with Applications (April 2026)

- Developed and implemented a finite volume scheme
- Performed numerical simulations in Python
- Analyzed convergence and stability

Preprint (Revision)
Finite Volume Scheme for the 1D Keller–Segel Chemotaxis Model using FASTEST Technique
Co-authors: A.C. Kengni Jotsa, J. Tagoudjeu, M. Zagour
Submitted to: Journal of Applied Mathematics and Computing (05/06/2026)

- Designed numerical scheme for 1D chemotaxis model
- Conducted error analysis and validation

WORK IN PROGRESS

- **Title**:"Mathematical analysis for finite Volume Scheme Based to FASTEST for Keller-Segel system apply to breast of cancer"
  Status: In progress

SCIENTIFIC EVENTS AND PUBLIC TALKS

- **A – PIC 2026 Summer School in Kigali:** 27/07/2026-07/08/2026
  role: Attending and given a talk on Algebraic cryptanalysis of coded-based cryptosystems
- **M2BE: Mathematical Modelling for Bioscience and Engineering** 07/07/2025-11/07/2025 at National Advanced School of Engineering of Yaounde 1
  role: Attending and given a talk on Numerical analysis of chemotaxis problems of Keller-Segel type
- Presentation of my Master Thesis at University of Maroua to hold a master's degree in Applied mathematics, 03 Oct. 2025
- Presentation of my Essay thesis Entitled *Algebraic Cryptanalysis of Code-Based Cryptography* at AIMS-Cameroon , 09 Jun 2026.

SKILLS

Programming: Python (NumPy, SciPy), R, SageMath, Scilab
Mathematical Tools: Numerical Simulation, PDE solvers
Other: Scientific computing, algorithm design

REFERENCES

- Dr.Antoine Celestin Kengni Jotsa
  Lecturer ,University of Maroua
  mailto:akengni60@gmail.com

- Dr. Franck Rivel Kamwa Djomou
  tutor ,AIMS-CAMEROON
  rivel.kamwa@aims-cameroon.org

- Prof. Pierre-Louis Cayrel
  Maitre de conference HC-HDR ,Directeur des etudes BUT2 GEII
  ,Laboratoire Hubert Curien
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